



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 121 : Structured Programming Language

Batch: 28th

Date: 07/12/2025

Midterm Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

- ✓ (a) Differentiate between Compiler and Interpreter. CLO1 2
- (b) Write the algorithm and draw the corresponding flowchart to determine the largest number among three given numbers (num1, num2 and num3). CLO2 4
2. (a) Why is the main() function essential in every C program? CLO4 2
- (b) Find errors, if any, in the following programs — CLO5 4×1=4
- i.

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", a, b);
    printf("The sum is: %d\n", a + b);
    return 0;
}
```
- ii.

```
#include <stdio.h>
INT MAIN() {
    char c = 'A'
    printf("The character is: %c\n", c);
    return 0;
}
```
- iii.

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", a, b);
    product = a * b;
    printf("The product is: %d", product);
    return 0;
}
```
- iv.

```
#include <stdio.h>
void main() {
    printf("Hello, World!");
    return 0;
}
```
- ✓ (a) What are the rules for naming an identifier in C? CLO4 2
- (b) Classify the following variable names as valid or invalid. For the invalid ones state the reason. CLO3 4×.5=2



- i. *Column-total*
 - ii. *1st_row*
 - iii. *Maximum*
 - iv. *return*
- (c) How does the double data type differ from float in terms of precision and storage? CLO3 2
4. (a) Define Operator Precedence and Associativity. CLO3 1
- (b) Explain how the conditional operator works in C programming. CLO4 2
- (c) Find out the output of the following code snippet — CLO4 3
- ```
#include <stdio.h>
main() {
 int a, b, c, d;
 a = 11;
 b = 7;
 c = --a - b;
 printf("a = %d b = %d c = %d\n", a, b, c);
 d = b-- * ++a;
 printf("a = %d b = %d d = %d\n", a, b, d);
 printf("a += b = %d\n", a+=b);
}
```
5. (a) State the differences between implicit and explicit type conversions with examples. CLO3 3
- (b) Explain how the scanf function can be used to test whether any errors occurred while reading input. CLO4 3
6. (a) What are character test functions? Provide some examples. CLO4 2
- (b) Determine the output for the following printf statements: CLO4 2
- ```
float num = 12.34567;
printf("%f\n", num);
printf("%.2f\n", num);
printf("%.3f\n", num);
printf("%.1f\n", num);
```
- (c) Write a C Program to check if a given character is vowel or not. CLO5 2
7. (a) Write a C program that implements a simple calculator using a switch-case statement. CLO5 3
- Requirements:
- i. Input two numbers (num1 and num2) using scanf.
 - ii. Input an operation (+, -, *, or /).
 - iii. Select the operation using a switch-case and perform the operation.
 - iv. Display the result.
 - v. Handle division by zero.
 - vi. Handle false input.
- (b) Write a C program to generate the Fibonacci series up to N terms. The value of N should be taken as input from the user. CLO5 3



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE 127: Electronic Devices and Circuits

Batch: 28th

Date: 11/12/2025

Midterm Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

- | | | | |
|--------|--|------|---|
| 1. (a) | Explain the band theory of solids. | CLO1 | 2 |
| (b) | Based on band theory, describe the electrical conductivity of different types of elements (conductors, semiconductors, insulators). | CLO1 | 4 |
| 2. (a) | Define the following terms:
i) Conduction band ii) Valence band iii) Forbidden energy gap | CLO1 | 3 |
| (b) | Describe how doping a semiconductor with a trivalent element increases its conductivity? | CLO1 | 3 |
| 3. (a) | Explain the outcome when one region of a silicon sample is doped with a trivalent impurity and another region is doped with a pentavalent impurity. | CLO1 | 2 |
| (b) | Describe the V-I characteristics of a PN junction diode in details highlighting the behavior under forward and reverse bias conditions. | CLO2 | 4 |
| 4. (a) | Define the following terms:
i) Quiescent point ii) Barrier voltage iii) Zener voltage | CLO2 | 3 |
| (b) | A PN junction diode is connected with its positive terminal to the positive supply and its negative terminal to the negative supply. Identify the type of biasing and briefly describe its effect. | CLO1 | 3 |
| 5. (a) | Define and describe the operation of a half-wave rectifier with appropriate circuit diagram. | CLO1 | 3 |
| (b) | Derive the expressions for the DC voltage and DC current of a half-wave rectifier. | CLO1 | 3 |

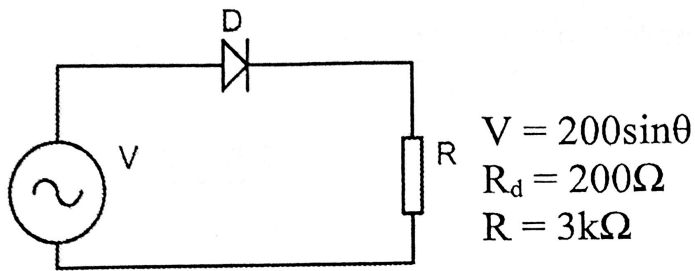
6. (a) Define the following terms:
 i) Full wave rectifier ii) DC voltage of a rectifier

CLO1 1+1=2

2+2=4

- (b) Observe the following circuit diagram and answer the following questions:

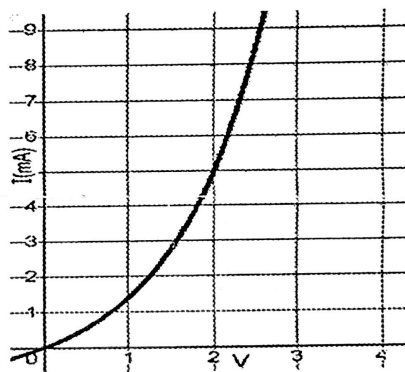
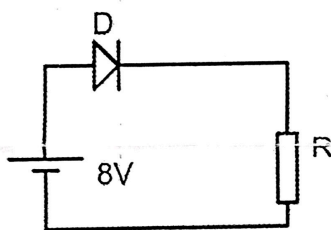
CLO2



- i) Calculate the dc voltage of the rectifier.
 ii) Calculate the efficiency of the given rectifier.

7. Observe the following circuit diagram and V-I characteristics and answer the following questions.

CLO2 2+3+1=6



$R = 1.2k\Omega$

Figure: V-I characteristics of diode D

- i) Find out the operating point.
 ii) Calculate the voltage drop and power dissipated across the load resistor.
 iii) Calculate the current flowing through the diode.



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 121

Course Title: Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Mid-Term Examination

Semester: Fall 2025

Date: 12/14/2025

Total Time: 1:30 Hours

Total Marks: 30

Answer any three(03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- | | | | | |
|---|-----|--|------|---|
| 1 | (a) | Define the following terms | CLO1 | 2 |
| | | i. Charge | | |
| | | ii. point charge | | |
| | | iii. conductor | | |
| | | iv. insulator | | |
| | (b) | State and explain Coulomb's law of electrostatics. | CLO2 | 4 |
| | (c) | The uranium nucleus contains a charge 92 times that of a proton. If a proton is shot at the nucleus, how large a repulsive force does the proton experience due to the nucleus when it is 10^{-11} m from the nucleus Centre? The nuclei of atoms are of the order of the 10^{-14} m in diameter, so the nucleus can be considered a point charge. | CLO3 | 4 |
| 2 | (a) | Define Electric field strength and lines of force. | CLO1 | 2 |
| | (b) | A charge $q_1 = (+1.0 \times 10^{-6} \text{ C})$ 10 cm from a charge $q_2 = (+2.0 \times 10^{-6} \text{ C})$. At what point on the line joining the two charges is the electric field strength zero? | CLO3 | 3 |
| | (c) | Show that the electric field due to an infinite line of a charge at a point is given by
$E = \lambda / 2\pi\epsilon_0 y$. Where λ is the linear charge density and y is the perpendicular distance of the point from the line. | CLO2 | 5 |
| 3 | (a) | Define electric flux. Give an expression for the flux in an electric field. | CLO1 | 3 |
| | (b) | An electric dipole is placed in a uniform external electric field E . Show that the work that must be done to change the orientation of the dipole in the field is given by $W = -P.E$ where P is the electric dipole moment. | CLO2 | 4 |



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- (c) An electric dipole consists of two opposite charges of magnitude $2.0 \times 10^{-6} \text{ C}$ separated by a distance $l=1.0 \text{ cm}$. It is placed in an external electric field of $2.0 \times 10^5 \text{ N/C}$. CLO3 3
- a) What maximum torque does the field exert on the dipole?
 - b) How much work must an external agent do to turn the dipole from its initial alignment given by $\theta = 0^\circ$ to final alignment $\theta = 90^\circ$?
- 4 (a) Find an expression for the potential at a point due to a dipole. CLO2 5
- (b) State and prove gauss's theorem. Show that Coulomb's law of electrostatics can be deduced from it. CLO2 5
- 5 (a) Derive an expression for the energy stored in the electric field of a charged capacitor. COL2 3
- (b) Explain the term capacitance. Calculate the capacitance of a cylindrical capacitor of length L formed by two coaxial cylinders of radii a and b COL2 4
- (c) How much charge is stored in a capacitor consisting of two concentric spheres of radii 30 cm and 31 cm if the potential difference is 300 V ? COL3 3



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT125 Course Title: Mathematics-II (Linear Algebra and Complex Variables)

Mid-Term Examination

Semester: Fall 2025

Date: 12/17/2025

Total Time: 1 Hour 30 minutes

Total Marks: 30

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

		CLOs	Marks
Q1.	(a) What do you mean by Hermitian and skew-Hermitian matrices?	CLO1	[2]
	(b) Prove that in a complex field, every square matrix can be uniquely expressed as the sum of a Hermitian matrix and a skew-Hermitian matrix.	CLO2	[4]
	(c) Justify whether the vector $v = (1, -2, 5)$ is a linear combination of the vectors $v_1 = (1, 1, 1)$, $v_2 = (1, 2, 3)$, $v_3 = (2, -1, 1)$.	CLO3	[4]
Q2.	(a) Define eigenvalue and eigenvector.	CLO1	[2]
	(b) Find the eigenvalues and corresponding eigenvectors for the matrix	CLO3	[8]
	$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$		
Q3.	(a) What do you mean by homogeneous and non-homogeneous systems of linear equations?	CLO1	[2]
	(b) Prove that every square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew symmetric matrix.	CLO2	[3]
	(c) For what values of μ , the following system of linear equations has a unique solution, more than one solution and no solution	CLO3	[5]
	$\left. \begin{aligned} x + y - z &= 1 \\ 2x + 3y + \mu z &= 3 \\ x + \mu y + 3z &= 2 \end{aligned} \right\}$		
Q4.	(a) Define vector subspace.	CLO1	[1]
	(b) Prove that the intersection of two subspaces of a vector space is also a subspace.	CLO2	[3]
	(c) Verify Cayley-Hamilton theorem and determine the inverse of the matrix	CLO3	[6]
	$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$		
Q5.	(a) Define characteristic matrix and characteristic equation.	CLO1	[1]
	(b) State and prove the Cayley-Hamilton theorem.	CLO2	[4]
	(c) Determine the inverse of the matrix by row canonical form:	CLO3	[5]
	$A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$		



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science and Engineering Department of Computer Science and Engineering

BSC (Honors) in CSE
ENG 127: English Skills
Mid-Term Examination
Fall 2025

Batch: 28th
December 09, 2025

2nd Semester
Time: 1 Hour 30 Minutes

Total Marks: 30

CLOs	Question/ Statement	Level	Marks
CLO1	Read the passage carefully and answer questions 1-20 from the passage. Each question carries 01 mark.	Understand	20×1=20

Urban Farming

In Paris, urban farmers are trying a soil-free approach to agriculture that uses less space and fewer resources. Could it help cities face the threats to our food supplies?

A

In Paris, a new style of urban farming is becoming popular. On the east side of the city, inside a small warehouse, a young entrepreneur named Quentin de Harenne is growing vegetables without using soil. "We use aeroponics," he says. "The plants hang in the air, and their roots get a fine mist of water and nutrients." This method has clear advantages. It uses 90% less water than conventional farming and can produce up to 400 times more food per square metre. De Harenne's company, **Agricool**, is one of several Paris start-ups working on soil-free, *vertical farming*. These farms grow crops in stacked layers inside controlled spaces such as warehouses or shipping containers.

B

Vertical farming is not only found in Paris. It has also been used in Japan and the United States. However, Paris is becoming a major centre for this industry in Europe. The city's mayor, Anne Hidalgo, wants to turn Paris into a "15-minute city," where people can easily reach work, shops and recreation on foot or by bicycle. Vertical farming supports this idea by producing food near where people live. "Our goal is to bring production and consumption closer," explains de Harenne. "In Paris, we grow lettuces that can be picked and delivered to restaurants within a few hours."

C

Agricool's warehouse in Saint-Denis clearly shows how this system works. In the 500-square-metre space, tall vertical tubes filled with lettuces, herbs and strawberries stretch up toward the ceiling. LED lights act like sunlight, and the temperature and humidity are carefully controlled. "We can grow all year round, no matter the weather," says de Harenne. The farm produces about 1,000 kilograms of vegetables and fruits every month, supplying around 20 nearby restaurants and shops. "Customers love the freshness and the idea that it is grown locally," he adds.

D

However, vertical farming also faces difficulties. The equipment is expensive to set up and consumes a lot of energy. "LED lights and climate control need much electricity," de Harenne admits. The company is trying to reduce this by using renewable energy. Some critics say vertical farms may even create more carbon emissions than traditional farms if the electricity comes from fossil fuels. However, supporters point out that transporting food long distances also harms the environment. "A lettuce usually travels over 2,000 kilometres before reaching a plate," says Dr Elena Rossi of the University of Paris. "Vertical farming can greatly reduce these food miles."

E

Another benefit is that vertical farms need far fewer pesticides. Outdoor farms often use chemicals to protect crops. Nevertheless, indoor farms face fewer pests, so chemicals are rarely needed. "We hardly use any pesticides," says de Harenne. "So our produce is cleaner and healthier." Vertical farms also save land, which is very important in crowded cities. "Paris has very little farmland left," Rossi explains. "Vertical farming helps us grow food in the city without taking away green spaces."

F

Even with these advantages, vertical farming is still a small industry. Only a small part of the food in Paris comes from these farms. "It won't replace traditional farming soon," says Rossi. "But it can support it, especially for expensive crops like herbs and leafy vegetables." De Harenne agrees. "We focus on products that need to be fresh and local, such as salads and microgreens. For crops like potatoes or grains, traditional farming is still better." The taste of vertically grown food is also debated. Some chefs enjoy its freshness, while others say soil-grown produce has a richer flavour. "It depends on personal taste," says de Harenne.

G

In the future, vertical farming could grow rapidly in Paris. The city government is helping with funding and research, and more farms are being planned. "We want Paris to be a leader in urban agriculture," says Mayor Hidalgo. "This is about food security, sustainability and community." For de Harenne, it is a long-held dream. "I always wanted farming to return to the city," he says. "Vertical farming lets us feed urban populations in a clean and efficient way." As cities around the world face rising populations and climate challenges, Paris's soil-free approach may become a model for others.

CLO5	TRUE	if the statement agrees with the information	Analyze	7×1=7
	FALSE	if the statement contradicts the information		
	NOT GIVEN	if there is no information on this		

1. Vertical farming can produce a huge amount of food in a small space.
2. Paris is the only city in the world where vertical farming has been attempted.
3. Vertical farming is more popular than traditional farming in Paris.
4. The Paris mayor supports vertical farming because she wants local food production.
5. The number of people employed by Agricool was around a thousand.
6. Vertical farms use far fewer pesticides than traditional outdoor farms.
7. De Harenne believes that vertical farms should completely replace traditional farms.

CLO2 Fill the blanks

Understand 7×1=7

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

Answers start from paragraph A.

8. Vertical farming allows crops to be grown in _____ layers.
9. Agricoool's warehouse is located in the suburb of _____.
10. The farm produces about _____ kilograms of vegetables and fruits each month.
11. Customers appreciate that the produce is _____ grown.
12. One advantage of vertical farming is that it saves _____, which is limited in cities.
13. Some chefs think vertically grown food lacks the _____ of soil-grown varieties.
14. Mayor Hidalgo supports vertical farming to improve food security, sustainability, and _____.

CLO5 Choose the correct letter A, B, C or D from the following questions.

Analyze 06×1=06

15. What method does Agricoool use to grow vegetables?

- A. Aeroponics B. Hydropower C. Soil layering D. Indoor composting

16. Why does Paris consider vertical farming useful?

- A. It increases tourism. B. It allows food to be grown close to residents.
C. It reduces the cost of rent in the city. D. It replaces outdoor parks.

17. What helps plants grow inside the vertical farm in Saint-Denis?

- A. Natural rainwater B. Outdoor sunlight C. Soil and fertilizer D. LED lights

18. What is one major criticism of vertical farms?

- A. They use a lot of electricity. B. They require too many workers.
C. They damage green parks. D. They spoil the taste of vegetables.

19. Why do vertical farms need fewer pesticides?

- A. They use stronger chemicals. B. The plants grow faster.
C. Indoor conditions limit pests. D. The farms are located far from cities.

20. According to Rossi, what role will vertical farming likely play in the future?

- A. It will replace all traditional farms. B. It will support traditional farming.
C. It will only be used for grains. D. It will be banned in cities.

CLO5 21. Write a Summary on the passage within 70 words

Understand 5×1=5

CLO1 22. What is scanning in reading? Mention at least three examples of scanning.

Define 5×1=5